LALIT SURAMPUDI

Phone: (847) 975-0153
Germantown, MD
Email: lsurampudi@gmail.com
LinkedIn: https://www.linkedin.com/in/lalitsurampudi/

SKILLS

- **Programming Languages:** Python, R, SQL.
- **Machine Learning:** Classification, regression, cluster analysis, dimensionality reduction, outlier analysis.
- **Libraries:** Pandas, NumPy, Matplotlib, Seaborn, scikit-learn, SciPy, TensorFlow, Keras.
- **Statistics:** Hypothesis testing, statistical significance and p-values, confidence intervals, sampling methods.

PROFESSIONAL EXPERIENCE

Data Scientist, *Mines Safety and Health Administration, Arlington, VA* Aug 2017 – Present

- Develop predictive models to forecast trends in mine production, injury rates and resource allocation for mine inspections across the US mining districts.
- Improve model prediction accuracy by **20%** using ensemble tree models, outlier analysis, and feature engineering.
- Lead efforts to build interactive visualization dashboards for MDRS application which resulted in increased user count by **30%** (www.msha.gov/microstrategy).
- Coordinate with organization strategy planning team to perform data discovery, data visualizations and identifying trends, and report actionable insights to business stakeholders.

Data Scientist, *DMI, Bethesda, MD* Apr 2017 – Jul 2017

- Lead development of data science applications from conceptualization to production using agile methodologies.
- Developed **NLP** based classification tool to reduce manual document review hours by **50%** by providing statistical overview of the document content.
- Built machine learning pipeline to automate data ingestion, feature extraction, model retraining, testing and deployment of model into production.

Sr. Statistical Analyst, *DMI, Mason, OH* Mar 2016 – Mar 2017

- Directed business analytics projects by enhancing statistical models of product assortment and identified key marketing channels to increase their gross margin by **14%**.
- Developed Marketing Mix Model which resulted in an increase of **5%** in ROI for a leading optical retail chain.
- Responsible for building data acquisition and **ETL** based systems to perform univariate correlation analysis for more than **1000** variables from **30** different datasets.

Computational Research Assistant, *Tulane University, New Orleans, LA* Jan 2010 – Aug 2015

- Applied machine learning including regression, gradient descent algorithms for energy minimization of molecular structures, principal component analysis and clustering methods on simulation data to study surfactant aggregation behavior.
- Developed scripts in R and Python to extract datasets of size **~30 GB** in size generated from computer-aided molecular simulations as well as scripts to automate processes, manage and schedule job submission on High Performance Computing (HPC) clusters.
- Performed statistical analysis of thermodynamic properties including accuracy, measures of central tendency, distribution of errors and confidence intervals.

EDUCATION

- Ph.D. Chemical and Biomolecular Engineering, Tulane University Aug 2015
- M.S. Chemical Engineering, University of Louisiana at Lafayette May 2009

ACADEMIC HONORS

- Distinguished graduate student award Dec 2014
- Outstanding graduate student teaching award Apr 2010
- IBM corporation fellowship in computational science Mar 2010